

Mixed and Longitudinal Models Group 11

final document

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```
load("Group_11.RData")
DF$sex <- factor(DF$sex, levels = c("male", "female"),
                 labels = c("Male", "Female"))

DF$smoking_status <- factor(DF$smoking_status, levels = c(0, 1, 2),
                           labels = c("Never", "Former", "Current"))
```

Declaration

We declare that all students in the group have contributed sufficiently to the study and that the work is original and not copied from elsewhere.

Signed:

#Introduction (10-15 lines)

The aim of our analysis is to understand the evolution of motor symptom severity over time in patients diagnosed with Parkinson's Disease, and the influence of patient characteristics such as sex, age at diagnosis, and specific lifestyle factors on this evolution. In this study, Unified Parkinson's Disease Rating Scale (UPDRS) Part III is used to quantify motor symptom severity. This score ranges from 0 to 108, where the higher value indicates a more severe motor impairment.

500 diagnosed patients were enrolled in the study between 1995 and 2011. Assessments were conducted on patients at baseline, and then annually for 7 consecutive years. Several variables were recorded for each patient, including dates of each UPDRS assessment, the corresponding UPDRS Part III score at each visit, gender, and age of the patient when diagnosed. Their smoking history was documented at baseline. Our focus in this report is on the influence of smoking status on the disease trajectory.

A challenge often encountered in observational studies is the presence of missing data. This can be due to factors such as patient withdrawal, relocation, or inability to perform assessments as the disease progresses. We have also analysed the pattern of the missing data, and assessed whether the Missing At Random (MAR) assumption is reasonable in our case.

The structure of our report is as follows:

Section 2 presents a descriptive analysis of the data; Section 3 presents the statistical analysis, including the final model and analysis results; The Appendix details the procedure and the statistical techniques we have used when selecting our final model, and how we have dealt with missing data.

#Descriptive Analysis (1-2 pages)

##2.1 Sample Characteristics

###Age Distribution

##	n	Missing	Mean	SD	Median	Q1	Q3	IQR	Min	Max
## 1	500	0	64.6414	13.4897	65.1	52.575	76.325	23.75	41.1	88

The dataset has patients with baseline ages ranging from 41.1 years to 88 years, and no implausible values were observed. Overall, the age distribution reflects a typical Parkinson's disease cohort, with most patients diagnosed in their 50s to 70s and a moderate spread across older age groups.

###Sex Distribution

##	sex	n	prop
## 1	Male	250	0.5
## 2	Female	250	0.5

The dataset contains a balanced sample of male and female patients.

Smoking Distribution

##	smoking_status	n	Percentage
## 1	Never	261	52.2
## 2	Former	141	28.2
## 3	Current	98	19.6

The majority of patients included in the dataset have never been smokers, slightly less than one third are former smokers, and approximately 20% are current smokers.

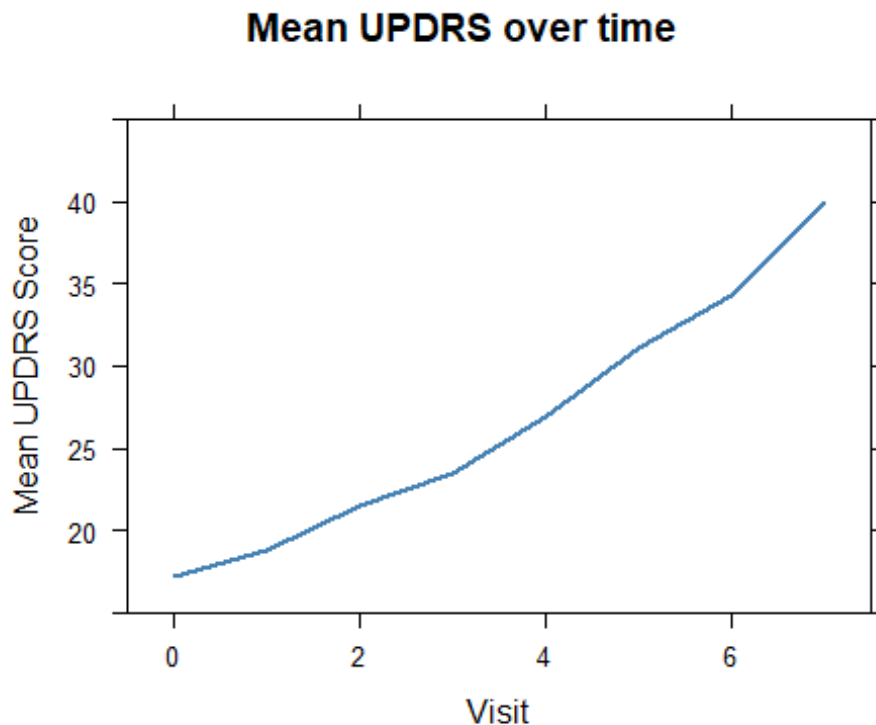
Missing Data

At baseline, all 500 patients were observed. At subsequent visits, observation counts ranged from 303 to 342, reflecting intermittent missingness over the follow-up period.

2.2 Longitudinal Outcome

In order to explore the longitudinal behaviour of UPDRS scores, we examined the mean structure, the variance structure, and the correlation structure over time.

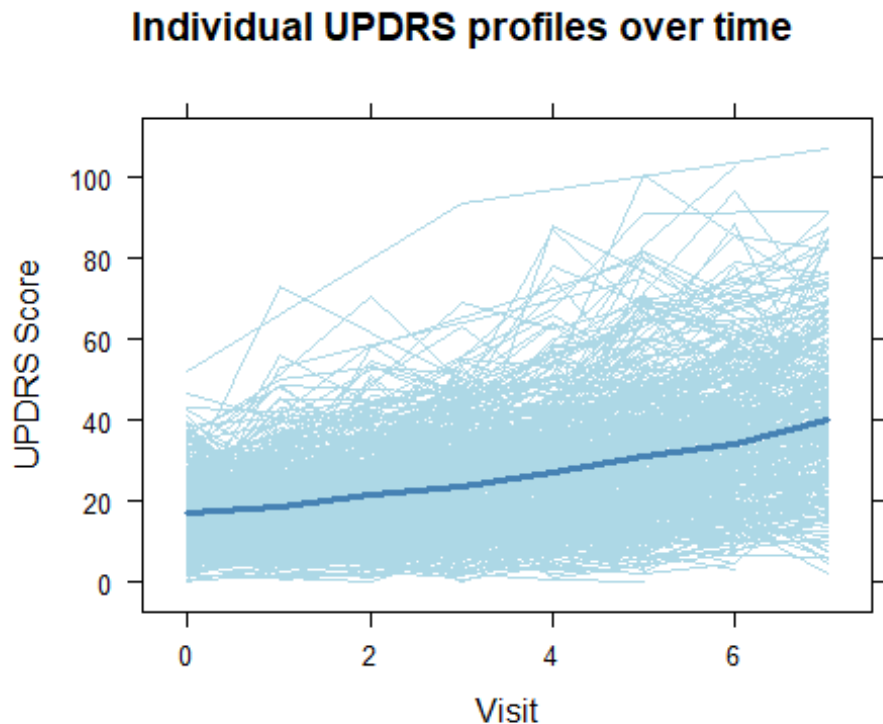
Mean Structure:



The mean UPDRS Part III score increases steadily across follow-up visits, indicating a clear progression of motor symptom severity over time. The mean UPDRS score increases from 17.24 at baseline to 39.99 at Visit 7. The trajectory appears approximately linear, although the slope

increases slightly in later follow-up visits. This may indicate a slight acceleration of symptom progression overtime.

###Variance Structure:



The spaghetti plot displays the longitudinal UPDRS trajectories for all patients over the 7-year follow-up period. Each light blue line represents the repeated measurements of a single patient, while the darker blue line corresponds to the average trajectory across all patients.

Overall, the average UPDRS score increases steadily over time, indicating a clear worsening of motor symptoms as the disease progresses. However, the individual trajectories reveal substantial heterogeneity between patients. In addition to the differences at baseline, patients also appear to differ in their rates of progression, suggesting that patients not only start at different levels of motor impairment, but may also deteriorate at different rates.

###Smoking and disease progression (Insert graph) At baseline, the mean UPDRS scores are relatively similar across the three groups. These small differences suggest that baseline motor symptom severity does not differ substantially by smoking status.

Over time, the trajectories for the three smoking groups are not perfectly parallel; former smokers show a steeper increase in later visits, while never smokers and current smokers show a more gradual progression. This lack of parallelism suggest the possible presence of an interaction between time and smoking status, meaning the effect of smoking status on UPDRS may change over time.

##Statistical Analysis (2-3 pages)

```
## Linear mixed-effects model fit by REML
##   Data: DF
##       AIC      BIC    logLik
##  21450.85 21521.9 -10713.42
##
## Random effects:
## Formula: ~Visit | id
## Structure: General positive-definite, Log-Cholesky parametrization
##              StdDev   Corr
## (Intercept) 4.410198 (Intr)
## Visit       1.885638 0.507
## Residual    9.726409
##
## Fixed effects:  UPDRS ~ Age + sex + Visit * smoking_status
##                  Value Std.Error   DF   t-value p-value
## (Intercept)      -0.9241595 1.8609664 2259  -0.496602  0.6195
## Age              0.2548847 0.0268528  495   9.491940  0.0000
## sexFemale        0.1059370 0.7227527  495   0.146574  0.8835
## Visit           2.9969081 0.1617279 2259 18.530559  0.0000
## smoking_statusFormer 1.0354207 0.8832240  495   1.172320  0.2416
## smoking_statusCurrent 0.6767738 0.9936519  495   0.681097  0.4961
## Visit:smoking_statusFormer 0.1577362 0.2757614 2259   0.572003  0.5674
## Visit:smoking_statusCurrent -0.1320765 0.3118966 2259  -0.423463  0.6720
## Correlation:
##                  (Intr) Age      sexFml Visit  smkn_F smkn_C
Vst:_F
## Age              -0.939
## sexFemale        -0.242  0.046
## Visit           -0.083  0.004 -0.001
## smoking_statusFormer -0.128 -0.040  0.000  0.167
## smoking_statusCurrent -0.168  0.014  0.044  0.149  0.309
## Visit:smoking_statusFormer 0.049 -0.004  0.005 -0.586 -0.290 -0.087
## Visit:smoking_statusCurrent 0.040  0.002 -0.002 -0.519 -0.087 -0.285
0.304
##
## Standardized Within-Group Residuals:
##           Min           Q1           Med           Q3           Max
## -2.77455726 -0.62436121 -0.09768422  0.54345959  4.54076002
##
## Number of Observations: 2762
## Number of Groups: 500
##
## Approximate 95% confidence intervals
##
## Fixed effects:
##                  lower      est.      upper
## (Intercept)      -4.5735419 -0.9241595  2.7252228
## Age              0.2021253  0.2548847  0.3076441
```

## sexFemale	-1.3141044	0.1059370	1.5259783
## Visit	2.6797574	2.9969081	3.3140589
## smoking_statusFormer	-0.6999095	1.0354207	2.7707508
## smoking_statusCurrent	-1.2755217	0.6767738	2.6290693
## Visit:smoking_statusFormer	-0.3830360	0.1577362	0.6985084
## Visit:smoking_statusCurrent	-0.7437104	-0.1320765	0.4795573

Visit: On average, UPDRS scores increase by 3.04 points per year of follow-up (95% CI [2.73,3.34]). This indicates a statistically significant worsening of motor symptoms over time.

Age: On average, UPDRS scores increase by 0.26 per a one year increase in the age of the participant at baseline (95% CI[0.20,0.31]). This indicates that older patients at diagnosis have significantly higher UPDRS scores on average.

Sex: Females have on average 0.12 points higher UPDRS scores than males (95% CI[-1.30,1.54]). There is no statistically significant difference in UPDRS scores between males and females after accounting for our other variables.

Smoking Status and Visit:Smoking status: There is no evidence that smoking history significantly influences either baseline motor symptom severity or the rate of progression over time (baseline: 95% CI [-0.50,1.38], progression rate:95% CI [-0.33,0.26]).

The between-patient standard deviation of random intercepts is 4.41, which tells us that patients differ substantially in their UPDRS levels at baseline. The between-patient standard deviation of random slopes is 1.88, which tells us that patients also differ in their rate of progression over time.

##Appendix (2-3 pages)